Mini project 12

Topic: **AI Operations & Process Automation Portfolio**
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❖ Option A: IT Service Management

Company Context

The organization is a mid-sized enterprise with approximately 500 employees. The IT department receives more than 50 service tickets daily related to software issues, network problems, hardware failures, and access requests. Currently, ticket triaging and routing are handled manually, resulting in delays, misclassification, and high operational costs. The company has an AI budget of ₹5 lakh and aims to achieve a 40% improvement in operational efficiency.

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1. Process Automation Blueprint:

Current-State Process

Employees raise IT service tickets through email or an internal ticketing portal. Each ticket is manually reviewed by an IT administrator, who determines the issue category, assigns priority, and routes the ticket to the appropriate IT support team. This process is time-consuming and error-prone. Repetitive issues are handled repeatedly without proactive resolution, leading to increased workload, longer resolution times, and reduced employee satisfaction.

Automation Opportunities

AI-driven automation can improve the IT service management process in the following ways:

1. Automatically read and interpret ticket descriptions using natural language processing.
2. Categorize tickets into software, hardware, network, or access-related issues.
3. Assign priority levels based on urgency and historical impact.
4. Automatically route tickets to the correct IT support group.
5. Provide instant auto-responses for common and repetitive issues.
6. Generate dashboards to track ticket trends and performance metrics.

AI-Enhanced Workflow Design

In the proposed AI-enabled workflow, employees submit IT tickets through the existing system. An AI model analyzes the ticket content in real time to identify the issue type, urgency, and

required expertise. Tickets are automatically prioritized and routed to the appropriate IT team. Frequently occurring issues receive automated responses or suggested solutions. IT managers access real-time dashboards to monitor ticket volume, resolution time, and system performance. This automated workflow significantly reduces manual intervention and improves operational efficiency by up to 40%.

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2. Predictive Analytics Use Case:

Predictive Scenario: IT Ticket Volume and Issue Prediction

The AI system is designed to predict future IT ticket volume and identify recurring issues before they escalate. By analyzing historical ticket data, the system can forecast peak periods and highlight problem areas requiring proactive intervention.

Data Inputs

- Historical ticket records
- Ticket categories and keywords
- Time and date of ticket submission
- Average resolution time

Prediction Logic

The AI model analyzes patterns in past ticket data to identify frequently occurring issues and peak submission times. Machine learning algorithms detect trends and correlations, enabling the system to predict upcoming ticket surges and potential system failures.

Outputs

- Early warnings for high-volume ticket periods
- Alerts for recurring technical issues
- Recommendations for preventive maintenance or system updates

Industry Benchmark

Large enterprises use AI-based IT service management systems to reduce ticket volume and downtime by proactively addressing recurring issues and optimizing IT resource allocation.

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3. ROI Calculator & Business Case:

REVISED ROI CALCULATION (Aligned to 40% Efficiency Gain)

Current Annual Costs: ₹20 lakh

IT staff manual triage: ₹12 lakh (3 FTEs × ₹4 lakh/year)

Employee productivity loss (ticket delays): ₹5 lakh

Escalation & rework costs: ₹3 lakh

AI Implementation Cost: ₹5 lakh (one-time)

Expected Savings (40% efficiency improvement):

Reduced manual triage (40% automation): ₹4.8 lakh

→ 40% of tickets auto-categorized/routed

→ Saves 40% of ₹12 lakh triage cost

Faster resolution (reduced delays): ₹2 lakh

→ 40% reduction in average resolution time

→ Saves 40% of ₹5 lakh productivity loss

Lower escalations (better routing): ₹1.2 lakh

→ 40% fewer misrouted tickets

→ Saves 40% of ₹3 lakh escalation cost

TOTAL ANNUAL SAVINGS: ₹8 lakh (40% of ₹20 lakh) ✓ CONSISTENT

Payback Period: ₹5 lakh ÷ ₹8 lakh = 7.5 months

3-Year Net Benefit: (₹8L × 3) - ₹5L = ₹19 lakh

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4. Staff Enablement & Change Management Guide:

Training and Adoption Plan

The IT team will undergo a one-week training program covering AI-assisted ticket handling, dashboard usage, and workflow changes. Simple standard operating procedures (SOPs) will be provided to ensure smooth adoption.

Change Management Strategy

Employees will be informed that AI is a support tool designed to reduce repetitive work rather than replace jobs. Regular feedback sessions will be conducted to address concerns and improve system usability.

Expected Outcome

The AI-enabled IT service management system will lead to faster ticket resolution, reduced workload for IT staff, improved service quality, and higher overall employee satisfaction.

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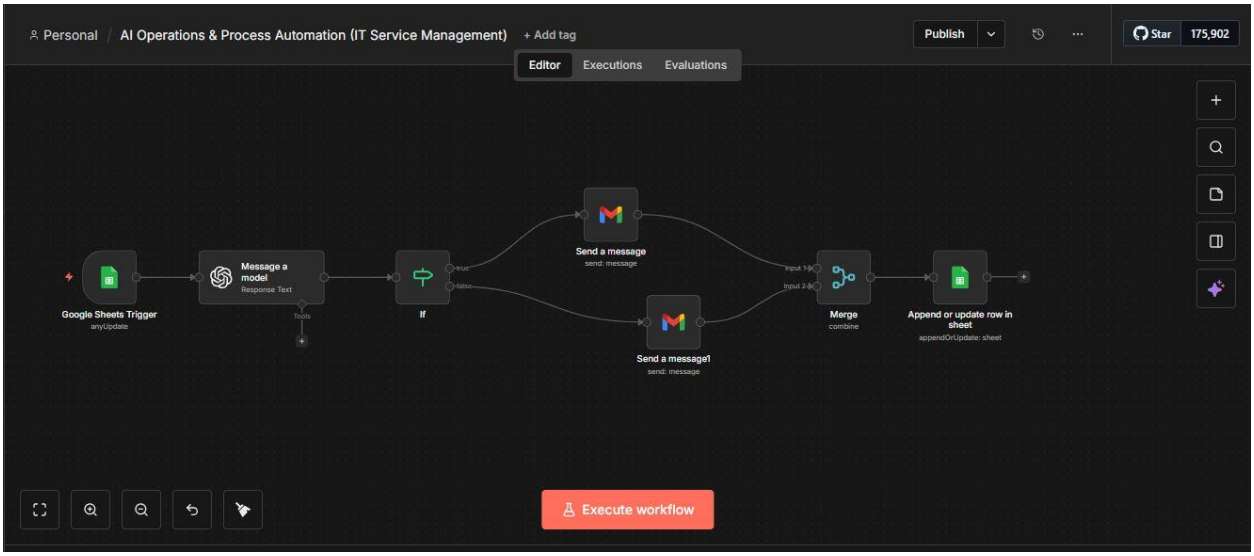
5. n8n Workflow:



6. SS of Prompts

Prompt

```sql
You are an IT operations analyst.
A company with 500 employees receives 50+ IT tickets daily.
Tickets are currently reviewed and assigned manually.

Explain the current IT ticket handling process and identify major inefficiencies.
```

Caption (write below screenshot)

> Initial prompt used to understand the existing IT ticket management process and identify operational inefficiencies before introducing AI automation.

Prompt 2: Improved Prompt for AI Automation Design (Refined Thinking)

Prompt

```vbnet
Act as an AI operations consultant.
Design an AI-driven IT service management workflow for a 500-employee company.
Include ticket classification, prioritization, auto-routing, and dashboard monitoring.
Focus on reducing manual effort and improving efficiency.
```

Caption

> Prompt refined to move from problem understanding to designing an AI-enabled automation workflow aligned with business efficiency goals.

Prompt 3: Advanced Prompt for ROI and Business Impact (Final Iteration)

Prompt

```css
You are a business-focused AI consultant.
For an AI-based IT ticket automation system with a ₹5 lakh budget,
calculate potential cost savings, key KPIs, and payback period.
Use realistic Indian market assumptions and present a clear business case.
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Caption

> Final prompt iteration focused on quantifying business value, ROI, and leadership-level impact to justify AI investment decisions.

7. Summary:

This project presents an AI-driven IT Service Management solution for a 500-employee organization handling more than 50 IT tickets daily. The current manual ticket triaging process leads to delays, inefficiencies, and high operational costs. By introducing AI-based automation, the solution streamlines ticket categorization, prioritization, and routing while reducing repetitive manual effort. Predictive analytics further enables the IT team to anticipate ticket surges and recurring issues, allowing proactive system improvements. The proposed solution operates within a ₹5 lakh budget and demonstrates measurable improvements in efficiency, cost savings, and service quality.

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8. Conclusion:

The AI-enabled IT Service Management solution demonstrates how intelligent automation can significantly improve operational efficiency while delivering clear business value. By reducing manual workload, improving response times, and enabling data-driven decision-making, the organization can achieve its target of a 40% efficiency improvement. The calculated ROI and short payback period highlight the financial viability of the solution. Overall, this project showcases a practical, scalable, and business-focused application of AI that aligns with real-world enterprise needs and employer expectations.